



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 • STANDARD 6.NOS.7

# Ordered Pairs in All Four Quadrants

Lesson 7-8 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can plot ordered pairs in all four quadrants of the coordinate plane.

**Language Objective:** I can explain my work using the words quadrant, negative coordinate, reflection, and axis.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Ordered Pairs in All Four Quadrants

Quadrant

Negative coordinate

Reflection

Axis

Integer



Tap any word to see what it means and a picture.

1 The  of a point's two coordinates decide which quadrant it lands in.

2 One of the four regions of the coordinate plane, numbered I to IV, is a

.

3 A coordinate less than zero, which places a point left of or below the origin, is a

.

4 A flip of a point over an axis is a .

5 One of the two number lines, the x-axis or y-axis, that form the grid is an

.

6

A whole number or its opposite is an  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How do four-quadrant coordinates help a city map locations in every direction from the center?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Ordered Pairs in All Four Quadrants** — Points written as  $(x, y)$  whose positive or negative coordinates place them in one of four quadrants.  
*Pares ordenados en los cuatro cuadrantes*  
— *Puntos escritos como  $(x, y)$  cuyas coordenadas positivas o negativas los ubican en uno de cuatro cuadrantes.*

☐ **Quadrant** — One of the four parts of a coordinate grid, numbered I to IV.  
*Cuadrante* — *Una de las cuatro partes de una cuadrícula de coordenadas, numeradas I a IV.*

☐ **Negative coordinate** — A coordinate less than zero. It means left or below the center point.  
*Coordenada negativa* — *Una coordenada menor que cero. Significa a la izquierda o debajo del centro.*

☐ **Reflection** — A flipped, mirror image across a line.  
*Reflexión* — *Una imagen reflejada al otro lado de una línea.*

☐ **Axis** — One of the two number lines that frame a graph: the x-axis runs across, the y-axis runs up and down.  
*Eje* — *Una de las dos rectas numéricas que enmarcan una gráfica: el eje x va horizontal y el eje y va vertical.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The library is in Quadrant \_\_\_\_ because its x is \_\_\_\_ and its y is \_\_\_\_ . The park is in Quadrant \_\_\_\_ because \_\_\_\_ . Four-quadrant coordinates help because \_\_\_\_ .**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_ .*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The signs of the coordinates tell me the quadrant because \_\_\_\_.

**Evidence:** Student matches sign combinations to quadrants (e.g.,  $(+,+)$  is I,  $(-,+)$  is II).

**Reasoning:** This evidence connects the definition of ordered pairs in all four quadrants to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The library is in Quadrant \_\_\_\_ because its x is \_\_\_\_ and its y is \_\_\_\_ . The park is in Quadrant \_\_\_\_ because \_\_\_\_ . Four-quadrant coordinates help because \_\_\_\_ .
- My answer is \_\_\_\_ .

*Mi respuesta es \_\_\_\_ .*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_ .  
*Mi afirmación es \_\_\_\_ .*
- I know because \_\_\_\_ .

*Lo sé porque \_\_\_\_ .*

- So \_\_\_\_ .

*Entonces \_\_\_\_ .*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Ordered Pairs in All Four Quadrants
2. **Notes 2:** Quadrant
3. **Notes 3:** Negative coordinate
4. **Notes 4:** Reflection
5. **Notes 5:** Axis
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Ordered Pairs in All Four Quadrants is matching each sign to the wrong axis — for example, seeing  $(-2, 5)$  and thinking "the  $x$  is negative so I move DOWN, and the  $y$  is positive so I move RIGHT," which lands the point in Quadrant IV. But the  $x$ -coordinate always controls left/right, never up/down: negative  $x$  means LEFT (not down), and positive  $y$  means UP (not right). So  $(-2, 5)$  is actually left 2, up 5 — Quadrant II. Before you submit, check that you used the  $x$ -coordinate for left/right and the  $y$ -coordinate for up/down, and never swapped them.*
8. **Try It 1:** C. Quadrant III — *A negative  $x$  means left and a negative  $y$  means down, so  $(-5, -2)$  is in Quadrant III, where both coordinates are negative.*
9. **Try It 2:** B.  $(-, +)$  — *Quadrant II is up and to the left of the origin, so  $x$  is negative and  $y$  is positive:  $(-, +)$ .*